

(IN)TRANSITIVITY: WHERE BRADLEY-TERRY FAILS TO MODEL HUMAN PREFERENCES

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ABSTRACT

Most preference-based alignment methods use Bradley-Terry (BT) preference models, which implicitly assume human preferences are consistent and transitive, i.e., that they can be collapsed onto a single scalar. In reality, however, heterogeneous human preferences can cause intransitivity through cyclic majority relations both at the user, as well as population level—a problem well-studied in social choice theory. While recent game-theoretic alignment methods propose cycle-aware objectives to handle this, whether such cycles matter empirically remains unclear. In this work, we answer this question by first auditing 9 widely used human-preference datasets and find that by construction, most lack the overlapping, repeated judgements required to analyze population-level intransitivity. We then construct INTRANSIT, a controlled dense preference dataset, through preference-completion methods that estimate the missing population-level preferences. [daniel: I’m not sure if these claims are really supported since we specifically chose responses in order to make cycles, and so many of these numbers depend on design choices. It’s not really clear what performance means here] We use this as a measurement tool for widely used preference models and show how performance degrades to near-chance agreement on intransitive samples. At the same time, the denser preference structure in the completed datasets contains useful signal in that preference models trained on this data perform as well as models trained on fully complete pairwise annotations. Finally, at inference time, using trained preference model outputs reduces regret and majority-rejected selections, and reinforcement-learning systems trained with preference models over this show improvements for certain preference objectives. Our results identify two challenges for preference alignment: collecting and measuring preferences to allow intransitive preferences to surface, as well as building objectives for the representation of these preferences without imposing a scalar ordering.

1 INTRODUCTION

Language model post-training predominantly relies on human preference judgements to align with humans. In reinforcement learning from human feedback (RLHF), human annotators compare candidate responses, and a dataset of such preference comparisons is used to learn a reward function that ranks responses in order of preference (Stiennon et al. (2020)). A standard formulation is the Bradley-Terry (BT) model that represents each response with a reward thus allowing for the modeling of pairwise preferences as a function of reward differences (Bradley & Terry (1952); Ouyang et al. (2022)). By construction, this model makes a consequential assumption: that the preferences being modeled follow a scalar and transitive ordering. Therefore, it assumes that if a response A is preferred to B and B to C , then A is preferred to C .

[daniel: It seems like an important aspect to BT we don’t discuss it that not only does it assume the underlying preferences are transitive (that’s actually a bit debatable as it does predict disagreement), but it’s that the preferences take a very specific form, that the fraction of the time a is preferred to b can be represented by rewards with $\text{logit}(\exp(r(a) - r(b)))$. Human preferences across a population need not take this form. When the underlying generation model does not match this structure, we know weird things can happen. This in the case of

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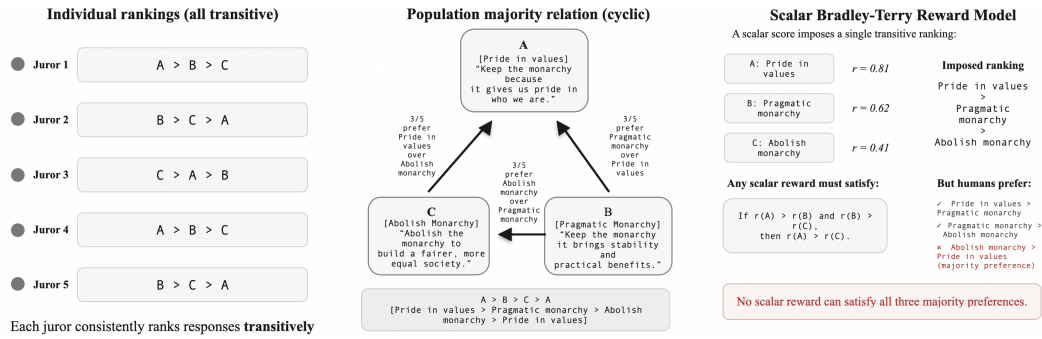


Figure 1: **Human judgements can produce cyclic population-level rankings** Five jurors from the Habermas study (Tessler et al. 2024) each rank three samples such that their majorities form a cycle ($A \succ B, B \succ C, C \succ A$). A scalar BT reward model on the right cannot correctly represent this.

cycles, yes, but even without cycles, it is possible for BT to prefer a non-Condorcet-winner even when when exists, because it is focused only on win rate.]

However, human preferences need not have this structure. [daniel: Given that our persona-based completion effectively assumes that people are transitive, and population disagreement is just due to disagreement across people, maybe we should downplay this a bit.] Irrationality in human decision-making has long been studied in individual choice (Tversky 1969; Regenwetter et al. 2011), where a single user can exhibit choice cycles in the presence of multiple conflicting attributes. At the population level, social choice theory has long established that cyclic majority relations can exist even when individual users exhibit transitivity (Arrow 1951; Sen 1966). Even in principle, a BT model cannot represent a violation of the transitive assumption leading to a consequence for language-model alignment: when models encounter a cycle, a scalar model must discard at least one relation in the cyclic graph. This concern has motivated a recent line of work proposing alignment methods that do not collapse preferences to scalar rewards, but are more *cycle-aware* through non-scalar objectives (Munos et al. 2023; Swamy et al. 2024; Rosset et al. 2024) [daniel: I would prefer a term other than cycle-aware. These methods don't fit to a population. Rather they are an alternate objective that (normatively) we think is more reasonable. It's not clear a priori if they do better on any of our actual metrics we are looking at (accuracy, majority consistency, etc). In fact, if our goals are those metrics, it probably makes sense to just optimize for them directly, which is is what things like IPO and alternative loss functions are already doing.]. However, an empirical question underlying these works remains unanswered, which is, do human preference datasets contain intransitive cycles? And if so, does this affect downstream language-model alignment performance? [daniel: Another important point is that it is not clear at all what we should do when there are cycles. It seems like most of our metrics is more about majority consistency, accuracy, calibration etc. These can be a problem even if there are no cycles in the dataset. I think this whole thing could generally be framed more around fitting well to the population.]

We answer this question in three stages. First, in §2 we ask whether existing preference datasets can reveal intransitivity in their annotations. [daniel: I again don't like the framing of "can reveal intransitivity" because it's not clear a priori that there necessarily is intransitivity (and I don't think we even at this point have enough statistical power to claim that there is). Rather, I think we can start this with the problem is that it is impossible to validate any of these claims about BT (or other method's) overall accuracy/majority consistency/calibration because the datasets are much too sparse. Therefore, we propose completing dataests in order to get some measure.] We audit 9 widely-used preference datasets and find that most are structurally unable to reveal cycles because they only satisfy disjoint sets of properties required to identify majority cycles. As a result, the apparent absence of reported cycles reflects a dataset annotation limitation rather than evidence of transitivity of human preferences. Second, we ask whether the missing preference structure can be recovered by "completing" these datasets in §3. We develop INTRANSIT, a dataset that serves as a controlled dense preference benchmark grounded in 1.3k real annotator profiles,

candidate responses and human judgements, and develop a population-level preference completion method to predict missing pairwise relations. We validate several completion methods on held-out human judgements and use the best-performing method that recovers unseen preferences well but tends to underestimate contested cyclic data, to construct the final dataset. Lastly, we evaluate the effect of these completed preferences on downstream alignment methods in §4. We fit BT models to different preference datasets, and show that the completed preferences provide useful training signal: preference reward models trained on completed datasets approach performance of exhaustive pairwise annotations. At inference-time, we see that using the completed preference predictions reduces both regret and majority-rejected selections. We also see mixed gains on post-training with this data; where some preference training objectives (e.g., DPO (Rafailov et al., 2023)) show improved performance, but others (IPO, KTO) show no improvements.

Put together, our results identify two important problems for preference alignment: measurement, since existing data collection regimes do not allow for intransitive preferences, as well as modeling, since preference objectives enforce scalar ordering. We show benefits of considering intransitive preferences that imply that alignment pipelines should test this assumption as well as preserve richer data when possible. We summarize our contributions as follows:

1. **A data-driven audit of intransitivity in preference datasets.** We evaluate several publicly available preference datasets for criteria required to identify population cycles and see that most, by construction, do not contain enough per-sample or per-rater annotations to reveal them.
2. **A methodology to recover missing preference structure.** We develop and validate preference-completion methods and use this to construct INTRANSIT. This lets us estimate missing preference comparisons to augment sparse preference datasets.
3. **Measuring model performance on intransitive data.** We evaluate several predominantly used preference models on completed datasets and see that models degrade sharply on cyclic samples, demonstrating the practical consequences of scalar preference reward structures.
4. **Using completed preference information downstream** We see that completed preferences improve training of preference reward models, have small post-training benefits, and improve inference-time serving to reduce regret and majority-rejected selections.

2 DOES INTRANSITIVITY EXIST IN PREFERENCE DATASETS?

2.1 PREFERENCE DATASETS

We audit 9 preference datasets used in language-model alignment. These datasets vary in how preferences are collected in several ways: the number of raters that judge each sample (single vs. repeated judgements), preservation of anonymized annotator identities (to track per-annotator judgements across samples), collecting pairwise judgements vs. picking one out of four samples and so on. We briefly outline the datasets below, and then compute properties of datasets and cycle existence across datasets in §2.2 and §2.3.

The datasets we consider fall into the following annotation structures: **Best-of-(k) datasets** include PRISM (Kirk et al., 2024), where participants evaluate up to four model responses to their own prompts and feedback is linked to persistent user profiles, and Community Alignment (CALign) (Zhang et al., 2026), which similarly elicits a choices from sets of four. **Pairwise datasets** include the Summarization-from-Human-Feedback (SHF) dataset (Stiennon et al., 2020), Anthropic HH-RLHF (Bai et al., 2022), MultiPref (Miranda et al., 2025), and LMSYS (Chiang et al., 2024), which provide comparisons between two responses at a time and therefore require repeated, overlapping comparisons to form observable cycles. HelpSteer2 (Wang et al., 2024) instead provides fine-grained attribute ratings of responses but lacks pairwise comparisons entirely. Finally, we consider two datasets that contain substantially denser population feedback: the Habermas dataset (Tessler et al., 2024) that provides full rankings from small groups deliberating over candidate statements, and OvertonBench (Poole-Dayana et al., 2026) that collects repeated 1–5 representation ratings from large participant samples across responses to value-laden questions. These latter datasets are smaller than the former, but provide unusually direct views of population-level preference structure. We

also considered other datasets with multiple annotators however some, such as DICES (Aroyo et al. 2023), Rastogi et al. (2026) contain safety ratings rather than preferences, and others such as Stanford Human Preferences (SHP) (Ethayarajh et al. (2022)) that are structurally similar to the datasets above and do not add new audit categories.

Dataset	Unique Raters	Raters / Sample	Total Samples	Annotation Format	Overlapping Comparisons	User IDs	Repeated Judgements	Cycles Possible	Cycle Rate
PRISM	1,309	1	8,011	choose 1 of 4	✓	✓	×	×	-
SHF	76	1	178,944	pairwise	✓	✓	≈	×	-
MultiPref	pseud.	2	5,323	pairwise	×	✓	✓	×	-
HelpSteer2	none	1	20,324	attribute ratings	×	×	×	×	-
CAlign	3,603	1	33,099	choose 1 of 4	≈	✓	≈	×	-
HH-RLHF	none	1	169,352	pairwise	×	×	×	×	-
LMSYS	pseud.	1	33,000	pairwise	≈	✓	×	≈	-
Habermas	5,734	5	7,340	full ranking	✓	✓	✓	✓	4.7%
OvertonBench	1,130	59	60	1-5 rating	✓	✓	✓	✓	8.3%
INTRANSIT	1,379	1,379	3,700	derived ranking	✓	✓	✓	✓	9.4%

Table 1: **Auditing Preference Datasets.** 9 preference datasets are audited for annotation structure required to observe cyclic majority relations. Only Habermas and OvertonBench provide repeated, overlapping, and user-preserving judgements [daniel: Why do we need user-preserving to detect cycles? Also maybe worth distinguishing observed cycles from true population cycles. Repeated questions give us a better estimate, but we do not know for sure that there actually is a population cycle, it could just be on this finite sample. Even a population perfectly fit to a BT model, we would expect on a finite-sized sampled dataset we would still observe cycles.] [roma: addressed!] that allow for cycle computation. We compare this to INTRANSIT that we construct in §3 and the properties it shows.

2.2 DETECTING CYCLES IN DATASETS

We represent the preferences associated with a prompt as a directed graph ($G = (V, E)$), where each node ($y_i \in V$) is a candidate response and an edge ($y_i \rightarrow y_j$) indicates that (y_i) is preferred to (y_j). We focus on the smallest such intransitive structure, a three-response cycle, ($y_i \succ y_j, y_j \succ y_k, y_k \succ y_i$) that cannot be represented by any scalar ordering over the three responses. We separate two sources of cyclic structure: *individual intransitivity*, where one annotator’s own judgments form a cycle, and *population intransitivity*, where the majorities form a cycle even though every annotator is transitive, as in the Condorcet paradox (Arrow 1951, Sen 1966). We focus on the latter.

In the constructed graph, we call a response triple an *observable cycle* if all three pairwise relations exist over the population. We treat missing edges as unobserved rather than transitive. For a given dataset, we therefore enumerate all observable triples and test whether their directed majority relation forms a cycle. We report this cycle rate as the fraction of eligible triples that are cyclic. When several annotators judge the same pair, we quantify uncertainty in each population relation through resampling, allowing us to distinguish cycles supported by stable majority preferences from those that may arise from finite-sample variation (see Appendix A.1 for resampling experiments and Appendix F.2 for further robustness checks).

2.3 AUDITING PREFERENCE DATASETS

In Table I we audit public preference datasets for the above criteria and reveal a big measurement limitation—that most datasets, given their annotation structures, cannot be evaluated for population cycles. Some datasets (e.g., PRISM) only contain one human judgement per response pair, while others (e.g., CommAlign) contain multiple judgements but lack repeated judgements from the same rater or sufficiently overlapping comparisons. Moreover, where some datasets retain anonymized annotator identification, most of them strip this information before release, making individual-level intransitivity for a rater impossible to distinguish from aggregation effects. We note two datasets that do contain dense human feedback, either through repeated or multiple annotators per sample: Habermas and OvertonBench containing a cycle rate of 4.7 and 8.3 respectively. This establishes that cycles can exist, when annotation structures allow them to be seen. However, these datasets were

constructed to elicit behaviors of deliberation and consensus, thus their cycle rates should not be interpreted as estimates of naturally occurring intransitivity, but only as evidence that intransitivity is detectable when the data supports it.

Our audit reveals a problem of measurement: that existing preference datasets do not contain overlapping or repeated judgments to determine whether their preferences are cyclic **[daniel: I think I would find this much more convincing if we present our other metrics (like accuracy/majority consistency etc) earlier, and claim that we are unable to measure these either. It is harder to motivate why cycles are intrinsically interesting, especially when these are just (potentially quite biased) estimates.]**. This motivates our question in §3 i.e., can the missing preferences in sparse datasets be estimated in order to “complete” these datasets and evaluate intransitivity?

3 COMPLETING SPARSE PREFERENCE DATASETS

3.1 PREFERENCE COMPLETION METHODS

Although the measurement problem we identify can be corrected with new data collection, we acknowledge that exhaustively collecting the type of data required is expensive. An exhaustive collection over n candidate responses requires $\binom{n}{2}$ total pairs, each ideally rated by multiple annotators. In this section we therefore ask the following question: can the missing structure be recovered by estimating the relations that don’t exist in current sparse preference datasets? We formulate this as a *preference completion* task. For a dataset with observed preferences (E_{obs}) over response pairs, we aim to fill in the missing preferences for every unobserved pair (y_i, y_j) by predicting the fraction of the population that prefers that ordering $[p_{ij} = P(y_i \succ y_j)]$. This is rooted in work on general preference modeling (Zhang et al., 2024b) and gives us a more robust target: rather than predicting the winning response, we predict the strength of the population disagreement itself.

Creating dense preference graphs for INTRANSIT To evaluate this methodology we require held-out dense preferences against which estimated preferences can be compared. We construct **[daniel: We should say synthetic somewhere]** INTRANSIT from 1,379 annotators profiles in PRISM, choosing this dataset because it records annotator’s relative assessments over three response attributes we focus on—helpfulness, safety and fluency. Therefore, an annotator’s attribute ordering over new responses can be estimated. However, the responses in this dataset are sparse, so we use PersonalLLM which consists of prompts and responses from eight frontier LLMs. We measure each candidate response along the response attributes and use annotator’s observed priority order to **[daniel: we should say somewhere that we assume that annotator utilities are linear in these attributes.]** estimate annotator-specific rankings. We then use rank-order centroid weights (Barron & Barrett, 1996) to convert each priority ordering into weights resulting in a ballot i.e., a complete response ranking for each annotator over candidate responses. **[daniel: I think we need to be more clear that we also generated responses specifically to get these cycles.]** See Appendix A.1 for details of construction and validation; one annotator’s ballot is worked through in Table 6. We also require cycles to survive resampling, specifically, two bootstraps over both the annotator population as well as the response-attribute measurements. The resulting dataset contains 19 certified majority cycles over 14 prompts and 26,201 annotator ballots. Notably, none of the 26,201 ballots is itself intransitive; hence showing how the cycles arise from aggregating heterogeneous but individually transitive preferences.

Estimating missing preferences for population-level completion One approach to complete missing preferences is to train a user simulator for each annotator (Wang et al., 2025). However, Table 1 shows how most preference datasets do not contain enough repeated judgements to do this, therefore we turn to modeling the population directly. We build on GPM (Zhang et al., 2024b), a model that learns K -dimensional representations of responses according to latent population components. For a response pair (y_i, y_j) , the model then predicts the population fraction $P(y_i \succ y_j)$ as the mixture of weighted components that prefer the first response. We adapt this into a population-based GPM (GPM-Pop) that predicts the share of a population preferring each response. On INTRANSIT, the population is the 1,379 annotators, each with their recorded weights; on public datasets without such records, the model learns a small number of latent groups. See Appendix C.3 for details of both. When fine-tuning on INTRANSIT we replace GPM’s scalar pair score with two heads: one

Table 2: **Completion Model Performance.** We report results on accuracy, error and cycle-accuracy on held-out human preferences across three datasets. GPM-Pop more accurately predicts preferences on contested subsets by a wide margin, but its calibration error remains imperfect, and no method can recover cycles in Habermas. [daniel: Why not report log loss? Isn’t that what GPM is trying to fit to?][daniel: Also how are these actually calculated on CAlign since there isn’t pairwise information?]

Dataset	Method	Acc(Decisive) ↑	Acc(Contested) ↑	MAE ↓	ECE ↓	Cycles
INTRANSIT	GPM-Pop	0.772	0.582	0.239	0.167	17/22
	PAL	0.771	0.485	0.237	0.102	9/22
	VPL	0.765	0.522	0.251	0.125	14/22
	Per-Fit	0.766	0.507	0.108	0.141	3/22
	GPM-Gen	0.658	0.425	0.135	0.029	6/22 [daniel:
Habermas	GPM-Pop	0.693	0.543	0.235	0.149	0/42
	Per-Fit	0.559	0.518	0.101	0.039	0/42
CAlign.	GPM-Pop	0.757	0.641	0.104	0.021	-
	Per-Fit	0.713	0.606	0.067	0.074	-

we only report false negatives for cycles should maybe also report false positives]

for K attributes that a response might have, and another for B groups that the population can be split into. The predicted fraction here therefore tells us about the share of the population whose group prefers one response over the other. During training, we upweight contested pairs three- to four-fold, since they are rare, and to deliberately optimize the model towards the contested regions that are sparse in human data. We report architecture, training, and dataset details for these models in Appendix C and evaluate performance against baselines that we describe in the next section.

3.2 EVALUATING COMPLETION METHODS

Data We evaluate the different completion methods on their ability to accurately predict human preferences on a held-out dataset. We hold out 28 prompts from INTRANSIT, each with all eight of its responses, yielding 784 response pairs and 1,568 triplets. For each pair, we compare the predicted population fraction with the ground truth, and check whether the method predicts the response that most of the 1,379 annotators prefer. We distinguish **decisive pairs**, where at least 65% of annotators preferred one response, from **contested pairs** where annotator division is split between 35% and 65%. We hypothesize that the latter is particularly important for intransitivity since cycles might arise at regions with substantial disagreement.

Metrics We evaluate completion performance at both the pairwise and graph levels with the following metrics. For held-out pairs, we compute majority accuracy (**Acc**) that tells us whether the model correctly predicted the response preferred by more than half the population and we report this on the **decisive** and **contested** pairs separately. We also report mean absolute error (**MAE**) between the predicted and ground-truth population preference fractions and calibration error (**ECE**) [daniel: How is calibration error defined? What binning do we use?]. Over the completed graph structures we evaluate the fraction of true cycles recovered among the triples that form real cycles in the dataset (**Cycles**) that tells us the precision of intransitive cycles being recovered.

Models We compare our population-based GPM (**GPM-Pop**) against two reward models that learn a personalized preference for each annotator. The Pluralistic Alignment model (**PAL**) (Chen et al., 2025) learns a shared set of reward functions and assigns each annotator individual non-negative weights. The Variational Preference Learning model (**VPL**) (Poddar et al., 2024) encodes each annotator’s recorded preferences into a latent vector, which is then used as an input to the reward model. We also fit a ridge regression on response embeddings to each annotator’s 25 comparisons (**Per-Fit**) and average the 1,379 fitted models into a predicted fraction per pair. We use the same baseline on Habermas and Community Alignment. Finally, we compare the general pre-trained GPM model (**GPM-General**). PAL, VPL and Per-Fit use 25 labels per annotator; GPM-Pop and GPM-General do not use any. See Appendix C for training configurations of these models.

Table 2 shows the difference in models performance on different parts of the dataset. On INTRANSIT all models except GPM-General perform similarly on decisive pairs reaching 77% majority-direction accuracy. On contested pairs, we see significant gains with GPM-Pop achieving a 58.2% accuracy compared to 48.5% and 52.2% for PAL and VPL, respectively. We also report the cycle-prediction accuracy on intransitive triples where GPM-Pop correctly predicts more cycles than other methods. Per-Fit and GPM-General have the lowest MAE on contested pairs simply because they predict fractions near 0.5—the same conservatism that reduces their accuracy. Our completion methods are therefore not to be interpreted as equivalent to individual human annotations, but rather an estimate of population-level preference structures; where the largest remaining errors are concentrated where the population is most divided. We also compare two other dense datasets and see similar trends: GPM-Pop outperforms Per-Fit on both decisive and contested samples, with the larger gain on decisive ones. On Habermas however, the GPM-Pop model fails to recover any of the true cycles in the dataset. Overall, this shows how population-level completion achieves sufficient accuracy on completion, and is able to recover harder, contested samples that contribute to dense preference structure. [daniel: I think this is too strong of a claim]

3.3 PROPERTIES OF COMPLETED DATASETS

Pairwise prediction accuracy, even if precise, does not tell us about structural differences in the completed datasets. We therefore compare graph-based properties of these datasets, along with distributional analyses of preferences that are estimated. This tells us whether completion methods that fill in missing edges significantly change properties of the datasets in comparison to the original human annotated versions. We consider 1,200 prompts with dense population graphs, of which 24.8% contain a cycle. When GPM-Pop predicts every pair of the same prompts from the response text, only 11.9% contain a cycle. Thus, completion makes populations look more transitive without softening how divided they are—the mean majority margin moves from 0.394 in the real data to 0.391 after completion. We are still able to recover intransitive samples in commonly contested cases. This distinction matters: completion methods underestimate intransitivity precisely for the reason that missing preferences hardest to predict are the contested preferences likely to cycle. Especially in contested regions, rather than a replacement of direct human measurement, we view this as a method for augmenting sparse data and demonstrate the usefulness of this downstream in the next section.

We also compare the distribution of the preference landscape before and after completion adapting the dataset cartography approach from Swayamdipta et al. (2020). For response pairs in the datasets, we measure the majority margin ($m_{ij} = |P(y_i \succ y_j) - 0.5|$) where larger values indicate annotator consensus and lower values close to 0 indicate disagreement. This tells us about the samples that fall into contested regions that models are uncertain about in the different datasets. We color each data sample accordingly to this difficulty estimate in Figure 2. We see that completion shifts the preference landscape towards lower variability and higher confidence, whereas also increasing coverage over contested regions. Together, with the reduced cycle rate above, this suggests a limitation for future work: although completion is expected to model the original data that does not have intransitivity, it does not cover the parts of the map that have contested judgements—which are the ones most relevant to intransitivity.

Figure 2: **Dataset map** for original (top) and completed (bottom) preferences. Points represent unseen response pairs, blue/red indicate contested/decisive preference sets. Completed preferences shift models towards towards high confidence, but contested relations are still rated as high variability.

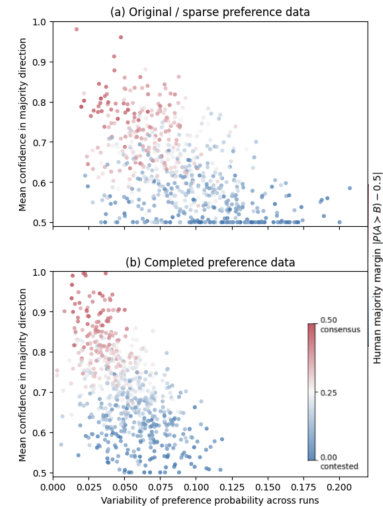


Table 3: **Preference models fail on intransitive samples, but training on completed data improves overall accuracy.** Comparing three model families on sparse vs. completed datasets shows improved overall accuracy, regret and transitive agreement. All scalar models degrade on intransitive subsets. A scalar ordering can satisfy at most two of three relations in a cycle (0.67) agreement. **[daniel: I think we need to seaparate these two tables. In my view, they are measuring very different things. For InTransit, the completed data is generated in the same way as the underlying held out data. E.g., it is asking if using more synthetic data is helpful for predicting heldout synthetic. It’s not too surprising that this is helpful given how sparse the data is. But that’s very different from Habermas to see if our synthetic procedure can help predict true ground truth human labels. That is very interesting if this truly happened. But we need to be very careful about what we are actually saying here to normalize these. I think at minimum we should be measuring and reporting the true bradley terry loss as well.]**

Objective	Trained on	INTRANSIT				Habermas			
		Acc. $\uparrow$	Trans./Intrans.	Regret $\downarrow$	ECE $\downarrow$	Acc. $\uparrow$	Trans./Intrans.	Regret $\downarrow$	ECE $\downarrow$
Bradley-Terry	sparse	0.671	0.673 / 0.529	0.307	0.149	0.661	0.662 / 0.542	0.368	0.013
Bradley-Terry	completed	0.719	0.721 / 0.503	0.272	0.047	0.688	0.694 / 0.519	0.349	0.090
IPO	sparse	0.628	0.630 / 0.509	0.350	0.036	0.644	0.644 / 0.528	0.378	0.102
IPO	completed	0.705	0.706 / 0.523	0.289	0.070	0.671	0.675 / 0.528	0.364	0.145
Rank-4 head	sparse	-	0.651 / 0.513	0.299	-	0.650	0.651 / 0.532	0.370	0.067
Rank-4 head	completed	-	0.701 / 0.539	0.291	-	0.682	0.688 / 0.528	0.352	0.135

4 USING COMPLETED PREFERENCES DOWNSTREAM FOR ALIGNMENT

4.1 HOW DO SCALAR PREFERENCE MODELS PERFORM ON CYCLES?

In this section, we validate the theoretical assumption we make about BT preference models: that their scalar reward modeling criteria does not allow handling of cyclic preferences. We compare three preference objectives: a BT model, IPO and a Rank-4 head objective.

All three use the same response embeddings. Both BT and IPO assign each response a single score but differ in their loss functions. BT directly fits the observed preference probability whereas IPO regresses the score difference to a predefined margin target. The Rank-4 head instead scores each response along four dimensions, with separate weighting for each value bloc. See Appendix [D.1](#) for details. **[daniel: So this can produce cycles right? It doesn’t have the same 0.67 cap is the others.]**

We hold out 300 of 1,200 INTRANSIT sets and 1,184 Habermas panels, whose questions are unseen in training, and report results of training each preference model on the original vs. completed versions of preferences in Table [3](#). We report **accuracy** against ground-truth preferences, as well as accuracy on transitive vs. intransitive (**Trans. / Intrans.**) subsets **[daniel: when we say subsets is this always triples?]**. We also report **regret** over samples and calibration error (ECE). When comparing prediction accuracy, we see that training on completed datasets improves performance when compared to the same model trained on the original dataset—e.g., BT accuracy increases from 67.1% to 71.9% on INTRANSIT and 66.1% to 68.8% on Habermas. IPO and Rank-4 show the same trends. Regret also improves for every model, while calibration improves only for BT on INTRANSIT. However, when comparing performance on transitive versus intransitive subsets, we see that performance drops sharply on intransitive i.e., cyclic data across the board. These results suggest two things. First, the performance drop on cyclic samples is expected, since additional supervision cannot significantly improve estimation of non-scalar rankings. However, second, we see that completing the missing preference relations contains useful supervision signal s.t., models trained on them do measurably better overall. Lastly, we also compare off-the-shelf preference reward models ranging in scale and architecture differences in Appendix Table [10](#). We see similar degradation of performance on intransitive samples across the reward models, consistent with our expected result.

4.2 POST-TRAINING ON COMPLETED PREFERENCES

We next ask whether the benefits of completed preference data extend from preference reward modeling to RLHF fine-tuning during post-training. We post-train a Qwen2.5 model ([Team, 2024](#))

Table 4: **Post-training gains from completed datasets depend on the optimization objective** Each policies response rank, as well as performance on dominated/contested subsets is reported. Completion improves DPO, but not IPO or KTO. Mean and S.D. over three seeds are computed.

Objective	Rank ↓	Dominated	Contested
DPO, raw labels	0.575 ± 0.016	93.8%	0.613
DPO, completed	0.544 ± 0.022	92.9%	0.587
IPO, raw labels	0.570 ± 0.021	94.5%	0.591
IPO, completed majorities	0.575 ± 0.035	94.9%	0.626
KTO, raw labels	0.572 ± 0.012	95.1%	0.609
KTO, completed majorities	0.596 ± 0.012	95.4%	0.628
Base model, untrained	0.625	98.0%	0.648
SPPO, completed fractions	0.583 ± 0.006	93.6%	0.618

Table 5: **Best-of- n ranking improves inference time performance.** On 300 held-out INTRANSIT samples, completion matches exhaustive pairwise supervision with dense samples. Selecting from the completed preferences with maximal lotteries gives the lowest regret and majority domination.

Selector	Best of 4		Best of 8	
	Regret	Dom.	Regret	Dom.
Reward model, raw (1 label)	0.345	52.7%	0.305	68.4%
Reward model, dense (28x)	0.317	49.0%	0.282	69.2%
Reward model, completed	0.316	49.5%	0.279	69.2%
Maximal lottery, completed	0.281	39.2%	0.219	54.0%
Oracle: population win rate	0.171	7.5%	0.073	10.7%

with the following preference optimization objectives. DPO (Rafailov et al., 2023), IPO (Azar et al., 2023) and KTO (Ethayarajh et al., 2024) take in pairwise comparisons in two settings: when trained on original data and completed data. An SPPO-style objective (Wu et al., 2025) operates directly against population preference probabilities, therefore only feasible with the completed data.

We evaluate each policy by the normalized rank [daniel: aren’t regret and rank measuring the same thing just inverses of each other?] of its response in each annotator’s ordering, the share of prompts where a majority prefers another response (**dominated**), and the share with no clear majority winner (**contested**) [daniel: what does contested mean in this context?]. In Table 4 we see the effect of completion depends on the optimization objective. DPO on completed data has the best mean rank of all the models, however IPO and KTO show slightly lower mean ranks when trained over the original datasets. SPPO obtains a weaker average rank than DPO but higher than the untrained base model. We speculate that these differences lie in the objectives: DPO’s implicit reward, compared to IPO’s fixed-margin regression. From this, we conclude that the benefit of completed preferences to post-training is highly dependent on the downstream objective; richer preference data does not alone guarantee a better policy, and methods that do not scalarize preference structures should be developed to see more benefits.

4.3 PRESERVING NON-SCALAR PREFERENCES AT INFERENCE TIME

In the above sections we demonstrate limitations of existing models that richer preference data cannot fully help overcome. Here, we ask how completed heterogeneous preferences can be preserved at inference-time to allow improvements to models. We analyze this through best-of- n serving, where an LLM produces n candidate responses and a preference reward model can be used to pick one by estimating preferences. For each of 300 held-out INTRANSIT prompts, we sample n of its eight responses as candidates, and each selector chooses one. The selectors are a Bradley-Terry reward model trained on one sampled human label per set (*raw*), on all 28 pairwise labels per set (*dense*), or on the completion of the raw labels (*completed*), and a maximal lottery (Maura-Rivero et al., 2025) over the completed fractions [daniel: Is this given the oracle pairwise proportions? This is a bit surprising to me, but I might be misunderstanding the experiment]. We report *regret*, how far down an annotator’s own ranking the returned response falls, from 0 (their favorite) to 1, averaged over all 1,379 annotators, and the share of draws in which a majority prefers another candidate. Lower is better on both metrics.

At best-of-8, a reward model trained on one raw label per set reaches a regret of 0.305; trained on the completion of those same labels it improves to 0.279, essentially matching the 0.282 of all 28 labels (Table 5). This confirms that the completed data with intransitive samples improves its average ranking, even though it does retain the same structural limitations of being scalarized. We then compare the maximal lottery approach applied to the entire population. This selects a distribution over responses that is not required to be ordered in a scalar manner. We see that best-of-8 here reduces regret to 0.219 giving the best performance, and also lowers majority-rejected selections to 54%. This highlights a new benefit and distinct way of using completed, intransitive preference data.

Instead of using them to train preference models that do indeed collapse intransitivity to a scalar reward, the non-scalar preferences improved best-of- n performance by preserving information that a scalar reward would otherwise discard.

5 DISCUSSION

Our results in this paper suggest two applications of intransitivity in preferences: as a **measurement** tool to test models’ ability to handle cyclic choices, and as a **representation** problem for preference alignment objectives. Our dataset audit shows how most existing preference datasets, by construction, are too sparse to show the existence of intransitivity. However, over the small number of cycles that do exist, we do in fact see that state-of-the-art preference models all fail to model these. Our proposed method to “complete” these cycles shows that we can recover some of the preference structure needed, and this proves useful for measurement of models, as well as downstream training. However, we acknowledge that this is only a complement, and not a replacement, for real human annotations. Fully understanding the extent of intransitivity—and its benefit to downstream training—will require a pipeline for dataset collection that fills in the currently missing parts of preference graphs. However, we do see that better preference data and representations solve some problems of current alignment systems, leading to better measures of regret on samples. More broadly, our results suggest a shift in data collection and preference alignment objectives in order to handle intransitivity. Instead of assuming preferences are scalar, and aggregating them into objectives, alignment pipelines can explore preserving intransitive preference structure in their objectives. This sort of alignment that treats disagreement as information to represent and reason over can lead to systems that are capable of adapting to diverse preferences in a more principled way.

6 RELATED WORK

Intransitivity in human and model preferences. Intransitivity in human choice has a long history in social-choice and decision theory. Individual-level cycles arise in the presence of multi-attribute tradeoffs of choices (Tversky 1969; Regenwetter et al. 2011; Butler & Pogrebná 2018) and population-level intransitivity can arise even in the presence of perfect individual-level transitivity (Arrow 1951; Sen 1966). Empirical work that studies intransitivity focuses on LLM judges and synthetic preference data (Tripathi et al. 2025; Xu et al. 2025; Hamilton et al. 2024) or strong group-level disagreement related to safety or value pluralism (Rastogi et al. 2026; Sorensen et al. 2025). Separate from all of these, our work asks whether existing preference datasets contain intransitivity, and if they do, how this preference structure affects downstream models using this data.

Modeling human preferences in populations Existing RLHF systems predominantly use scalar Bradley–Terry rewards (Bradley & Terry 1952; Stiennon et al. 2020), with newer methods relaxing this assumption to model pairwise or population-level objectives with game-theoretic objectives (Munos et al. 2023; Swamy et al. 2024; Rosset et al. 2024) or latent preference representations to model preferences (Poddar et al. 2024; Zhang et al. 2024a). Our work complements this, not by focusing on new objectives, but the interaction of intransitive data with these methods. We empirically evaluate the benefits of denser preference data from different modeling approaches, towards the goal of strategic language models that can represent complex human preferences (Goktas et al. 2025).

7 CONCLUSION

We explore the question of intransitivity in human preferences and whether scalar assumptions made by preference models can be improved upon. We audit datasets to evaluate presence of intransitivity, propose a preference-completion method to recover incomplete preferences, and show how scalar models, predictably, fail on cyclic preferences. Our experiments on preference learning and downstream RLHF fine-tuning show how completed datasets can improve performance in some settings, although completion cannot replace human judgements in contested samples. We discuss the implications of these findings for future preference data collections, and stress that alignments systems should not assume transitivity as-is, but measure it empirically, and develop objectives to better handle it for more aligned systems.

AI USE STATEMENT

In this work, we used generative AI tools to aid with generation of INTRANSIT (namely, measuring response attributes and writing responses to specified attribute profiles), code implementation (for some experiments), and feedback on experimental design (for some experiments). We have not used generative AI tools for the following tasks, which are not applicable to this work: development of theoretical models or conceptual frameworks, formulation of mathematical claims, provision of critical ingredients for proving mathematical claims, assistance in the writing of proofs, refinement of hypotheses, translation assistance, and support with qualitative or thematic data analysis. Additionally, we used generative AI tools for help with formatting of LaTeX tables and figures and revisions to paper writing. We have reviewed all AI-assisted work. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

ETHICS STATEMENT

Even though we release a new dataset with human preferences, which is an area prone to biases and concerns with regard to safe treatment and representation of users across different backgrounds, we strive to prioritize experiments that, in fact, highlight how minority preferences might get washed away by existing methods. In terms of the data collection, we do not collect any new human data, but extend existing datasets. Any considerations pertaining to the data collection process itself should be directed at the authors of the respective publications (Kirk et al., 2024; Zollo et al., 2025).

REPRODUCIBILITY STATEMENT

The code, data, and weights for the project are fully open-sourced at [anonymized](#)

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